



BIOLOGY CH.11 — ENDOCRINE & EXOCRINE SYSTEM — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway



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2. Endocrine Glands - Master Gland (Pituitary)
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7. Other Glands & Exocrine Functions
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1. Exocrine Glands & Digestive Enzymes

- Bile salts (liver/pituitary द्वारा produced) fat के emulsification में मदद करते हैं - fat को छोटे droplets में तोड़ते हैं ताकि enzymes digest कर सकें। [Q655]
- Trypsin (small intestine): digestion में मदद करता है। HCl (stomach): acidic environment और microbes को kill करता है। Pepsin (1836 में Theodore Schwann ने discover किया): endopeptidase enzyme जो proteins को peptides में तोड़ता है। [Q655]
- Salivary glands exocrine glands हैं - mouth में होते हैं, hundreds of small glands tongue/palate/lips/cheeks में। Ductal system में substances secrete करते हैं। [Q656]

2. Endocrine Glands - Master Gland (Pituitary)

- Pituitary gland (Hypophysis) "Master gland" कहलाती है - hypothalamus से thin stalk (pituitary stalk/infundibulum) द्वारा connected है। Pea-sized, oval, skull के andar. Anterior + Posterior lobe होते हैं। [Q657, Q664]
- Pituitary vital body functions, growth (height), और sexual development regulate करती है। Dwarfism इसके dysfunction से होता है। [Q658]
- Pituitary hormones: Growth Hormone (GH), Prolactin (PRL), Luteinizing Hormone (LH), Follicle-Stimulating Hormone (FSH), Thyroid-Stimulating Hormone (TSH), Adrenocorticotrophic Hormone (ACTH), Antidiuretic Hormone (ADH), Oxytocin. [Q664]
- Key gland locations: Hypothalamus और Pineal skull के andar, Thyroid neck के base पर, Thymus upper chest में, Adrenals kidneys के upar. [Q662]

3. Thyroid Gland

- Thyroid: butterfly-shaped, throat के base पर। Thyroxine (T4) और Triiodothyronine (T3) produce करती है जो growth, development, और overall metabolism को control करते हैं। [Q658, Q661]

4. Adrenal Glands

- Adrenal glands: kidneys ke upar hote hain. Cortisol, aldosterone, adrenaline produce karte hain jo blood pressure control karte hain. [Q659]
- Adrenal gland "Fight or Flight" response ke liye body ko prepare karti hai - isliye ise "Emergency hormone" bhi kehte hain. [Q660]

5. Pancreas - Dual Function

- Pancreas: Exocrine (digestion) + Endocrine (blood sugar regulation) dono functions. Abdomen mein hota hai. [Q658, Q661]
- Insulin (Beta cells se) aur Glucagon (Alpha cells se) secrete karta hai jo glucose regulate karte hain aur energy release ki rate control karte hain. [Q661, Q663]
- Blood sugar rise hone par Pancreas ki Beta cells detect karti hain aur Insulin secrete karti hain sugar kam karne ke liye. [Q663]
- Hyperinsulinemia = zyada insulin. Diabetes = inadequate insulin action ya production. [Q663]

6. Pineal Gland

- Pineal gland: small, pea-sized, brain ke third ventricle ke peeche. Melatonin secrete karti hai jo circadian rhythm/sleep regulate karta hai. Light-dark cycle ki information convey karti hai. [Q660, Q661]

7. Other Glands & Exocrine Functions

- Sebaceous glands (skin/hair follicles mein) sebum (oil) produce karti hain jo skin ki moisture maintain rakhti hain. Sweat glands sweat secrete karti hain. Puberty mein inka secretion increase hota hai. [Q666]
- Sebum mein triglycerides, wax esters, squalene hote hain. [Q666]

8. Liver & Glucose Metabolism

- Liver mein Gluconeogenesis hota hai - lactate (lactic acid) → pyruvate → glucose. Is glucose ko blood se muscles ko supply kiya jata hai. [Q667]

9. Excretory System Basics

- Kidney ki structural/functional unit Nephron hai - Neuron brain ki cell hai, Cytone/ Ketone galat options hain. [Q668]
- Urea liver se kidneys ke through excrete hota hai. [Q669]

- Insufficient insulin se Diabetes (Madhumeh) hota hai - high blood sugar. [Q670]
- Pancreatic duct: pancreas se duodenum tak digestive juices le jata hai. [Q671]
- ⚠ Trap: Endocrine glands ko "Ductless glands" bhi kehte hain kyunki ye bina duct ke directly bloodstream mein hormones secrete karte hain. [Q665]